Fiscal Year:	FY 2013	**Task Last Updated:**	FY 05/21/2013
PI Name:	Held, Kathryn Ph.D.		
Project Title:	3D Tissue Models for Study of Intercellular Signaling Stimulated by High Energy Particles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	kheld@partners.org	**Fax:**	FY 617-724-8320
PI Organization Type:	NON-PROFIT	**Phone:**	617-726-8161
Organization Name:	Massachusettes General Hospital		
PI Address 1:	Radiation Oncology		
PI Address 2:	55 Fruit Street		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02114-2696	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Radiobiology NNJ11ZSA001N
Start Date:	01/01/2012	**End Date:**	02/28/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 2/28/2013 per NSSC information (Ed., 10/25/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX12AB61G		
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Performance Goal Text:			
Task Description:	In space, astronauts are exposed to a complex, unique radiation environment including energetic protons and charged particles from galactic cosmic rays and solar particle events. Although these radiations can result in significant cumulative doses to astronauts during long-duration space flights, most exposures occur at low fluences, so that particle traversals through an astronaut's body are well separated in tissue location and time. Under such conditions, bystander effects, where unirradiated cells express biological changes in response to signals from irradiated cells, have increased significance. Our overall hypothesis is that increased oxidative stress caused by charged particle-stimulated intercellular signaling-- bystander effects-- in cells and tissues creates a microenvironment conducive to altered carcinogenic processes; responses can be triggered at very low doses, with the lowest effective dose depending on radiation type, but the magnitude of the response being radiation quality-independent. Our previous studies showed that bystander responses can be induced in unirradiated cells in a 3D skin model, including after low fluences of protons or iron ions, and that in 3D both irradiated and bystander cells are more resistant to induction of DNA damage than cells in monolayer. Different epithelial tissues can differ in their 3D architecture and, therefore, might be expected to have different radiation and bystander responses. Thus, in this project we proposed to compare bystander signaling in 3D models of colon with our previous data for the skin epithelial model. Our Aims were: 1. Develop a 3D model of colonic epithelium for testing effects of low fluences of charged particles in comparison with the responses in our previously developed 3D skin model. 2. Quantify DNA damage, apoptosis, differentiation and proliferation in the 3D model of colonic epithelium exposed to low fluences of charged-particles. Determine whether the bystander response in the colon model has the same radiation dose and LET dependence as the skin model, and whether reactive oxygen species (ROS) play a role in the intercellular signaling.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	The information gained in this project on cellular damage, intercellular signaling, and molecular processes that may be involved in development of cancer could also be important for increasing understanding of the potential for development of second cancers in cancer patients treated with photons, protons, or heavier ions such as carbon. Such information could ultimately lead to medical countermeasures that might be used in cancer patients to decrease incidence of or prevent treatment-induced second cancers.		
	SUPPLEMENTAL REPORTING FOR FINAL REPORT--MAY 2013: Background and brief description of project In space, astronauts are exposed to a complex and unique radiation environment including energetic protons and high atomic number, high energy ions from galactic cosmic rays and solar particle events. Although these radiations can result in significant cumulative doses to astronauts during long-duration space flights, exposures occur at low fluences and rates, so that particle traversals through an astronaut's body are well separated in tissue location and time. Under such conditions, bystander responses have increased biological significance because the fraction of cells receiving dose (from particle traversals or associated delta-rays) will be small compared to the proportion of cells that have the potential to exhibit bystander responses. Intercellular signaling, particularly between epithelial/pre-neoplastic/neoplastic cells and their stroma, are critical to the initiation and progression of tumorigenesis or to its prevention. However, the role of ionizing radiation in altering such intercellular signaling and/or initiating new signaling processes is not clear. Furthermore, the relationships among radiation quality, radiation-induced and bystander DNA damage, and carcinogenesis processes are not clear, and the role of tissue 3D architecture in intercellular communication affecting carcinogenesis, although clearly important, remains to be elucidated in the context of space-relevant radiation. During studies funded by a previous NASA grant, we developed and used a 3D skin model. Those studies showed that bystander responses can be induced in unirradiated cells in this 3D model, including after low fluences of protons or iron ions, and that in 3D both irradiated and bystander cells are more resistant to induction of DNA damage than cells in monolayer. Different epithelial tissues can differ in their 3D architecture and, therefore, might be expected to have different radiation and bystander responses. Thus, in this project we proposed to compare bystander signaling in 3D models of colon with our previous data for the skin epithelial model. We chose colon epithelium for this mechanistic comparison because colon is one of the tissue sites with the largest cancer risks, and results with this model could complement other NASA-funded studies on charged particle-induced colorectal cancers in animal models. Our Aims were: 1. Develop a 3D model of colonic epithelium for testing effects of low fluences of charged particles in comparison with the responses in our previously developed 3D skin model. 2. Quantify DNA damage, apoptosis, differentiation and proliferation in the 3D model of colonic epithelium exposed to low fluences of charged-particles. Determine whether the bystander response in the colon model has the same radiation dose and LET dependence as the skin model, and whether reactive oxygen species (ROS) play a role in the intercellular signaling.		

Task Progress:	Summary of research accomplishments during the final reporting period We used the Caco2 colon epithelial cancer cell line for initial studies since this line has been used as a model for colon epithelium in published literature, including one radiation study. In initial studies with the cells in monolayer, we used our transwell insert co-culture system to assess DNA damage in both irradiated cells and unirradiated bystander cells sharing medium with the irradiated cells. Our data show induction of micronuclei (MN; surrogate for chromosome aberrations) in Caco2 cells 72 h after irradiation with X-rays, 1 GeV protons or 1 GeV/n silicon or iron ions. In all cases, there is a large, radiation dose-dependent increase in MN in irradiated cells, but importantly, there is up to a 3-fold increase in the fraction of unirradiated bystander cells having MN after co-culture with irradiated cells. Additional studies using induction of foci of the DNA repair-related protein 53BP1, a surrogate for DNA double-strand breaks, as an endpoint have been done in irradiated and bystander Caco2 cells after exposure to X-rays, protons, and iron ions. With all radiation types, in the irradiated population, the fraction of cells with foci is substantially increased at 1 h, is maximum 5 h after irradiation, then decreases substantially by 24 h. In the bystander cell population, the increase in fraction of cells with foci is slower, not being maximal until 12 h after irradiation and the start of co-culture, and still being elevated at 24 h. The maximum fraction of bystander cells with increased level of foci is about 3-fold higher than in untreated cells, similar to the magnitude of bystander effect seen using MN as endpoint. This delayed development and removal of damage in bystander cells compared to irradiated cells is similar to the responses we have seen in the 3D skin model. We have successfully established the 3D colon model with the CaCo2 cells as the epithelium and normal human fibroblasts as the stromal cells embedded in collagen. We have quantified 53BP1 foci in both 1 GeV proton and X-irradiated and bystander fibroblasts and the Caco2 cells in the 3D colon model. Comparing the 2D and 3D data in the CaCo2 cells suggests several findings: First, in 3D there are fewer foci per cell, including in untreated samples, so a cut-off of only 1 focus per cell for analysis was used in 3D compared to 5 foci per cell minimum for scoring in 2D. If the 5 foci per cell cut-off had been used in the 3D samples, the fraction of cells with damage would be much lower in 3D, consistent with our finding with the skin model (Lumpkins et al., in preparation; and described in previous reports). Second, the time courses for development and disappearance of foci in 3D are similar to those in 2D, with cells in both architectures showing the delay in bystander responses compared to responses in irradiated cells. We have extended the model to use normal human colon epithelial cells (FHC cells; from ATCC) in both 2D and 3D studies. The early data have some scatter, but the trends show that in monolayer, 53BP1 foci appear to be maximal at 5 h after irradiation in both irradiated and bystander FHC cells, but replicate experiments are needed to confirm this finding. Furthermore, although these cells grow very slowly, we have been successful in producing 3D constructs in which the FHC cells form a single epithelial layer on the basement material containing fibroblasts, consistent with the single-layered colon epithelium. ANNUAL REPORTING PROVIDED OCTOBER 2012: Background and brief description of project: In space, astronauts are exposed to a complex and unique radiation environment including energetic protons and high atomic number, high energy ions from galactic cosmic rays and solar particle events. Although these radiations can result in significant cumulative doses to astronauts during long-duration space flights, exposures occur at low fluences and rates, so that particle traversals through an astronaut's body are well separated in tissue location and time. Under such conditions, bystander responses have increased biological significance because the fraction of cells receiving dose (from particle traversals or associated delta-rays) will be small compared to the proportion of cells that have the potential to exhibit bystander responses. 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Develop a 3D model of colonic epithelium for testing effects of low fluences of charged particles in comparison with the responses in our previously developed 3D skin model. 2. Quantify DNA damage, apoptosis, differentiation and proliferation in the 3D model of colonic epithelium exposed to low fluences of charged-particles. Determine whether the bystander response in the colon model has the same radiation dose and LET dependence as the skin model, and whether reactive oxygen species (ROS) play a role in the intercellular signaling. Summary of research accomplishments during the reporting period: Our plan had been to develop a 3D model of colonic epithelium using human fibroblasts embedded in collagen and over-layered by normal colon epithelial cells. There are few available human colon epithelial cells, but we had planned to use the NMC460 cells from INCELL Corporation. Unfortunately, when we attempted to order the cells, INCELL informed us that both that cell line and the other colon epithelial line they advertise were temporarily unavailable. Therefore, we chose to use the Caco2 colon cancer cell line for initial studies since this line has been used as a model for colon epithelium in published literature, including one radiation study. In initial studies with the cells in monolayer, we used our transwell insert co-culture system to assess DNA damage in both irradiated cells and unirradiated bystander cells sharing medium with the irradiated cells. Our data to date show induction of micronuclei (MN; surrogate for chromosome aberrations) in Caco2 cells 72 h after irradiation with X-rays, 1 GeV protons, or 1 GeV/n iron ions. In all cases, there is a large, radiation dose-dependent increase in MN in irradiated cells, but importantly, there is up to a 3-fold increase in the fraction of unirradiated bystander cells having MN after co-culture with irradiated cells. Additional studies using induction of foci of the DNA repair-related protein 53BP1, a surrogate for DNA double-strand breaks, as an endpoint have been done in irradiated and bystander Caco2 cells after X-ray exposure. In the irradiated population, the fraction of cells with foci is substantially increased at 1 h, is maximum 5 h after irradiation, then decreases substantially by 24 h. In the bystander cell population, the increase in fraction of cells with foci is slower, not being maximal until 12 h after irradiation and the start of co-culture, and still being elevated at 24 h. 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If the 5 foci per cell cut-off had been used in the 3D samples, the fraction of cells with damage would be much lower in 3D, consistent with our finding with the skin model (Lumpkins et al., in preparation). Second, the time courses for development and disappearance of foci in 3D are similar to those in 2D, with cells in both architectures showing the delay in bystander responses compared to responses in irradiated cells. These data are all from single experiments. Repeats are now in progress. Furthermore, irradiations of 3D colon epithelial model samples with iron ions and protons were just done at NSRL in early October, and have not yet been analyzed. 3D samples will be irradiated with silicon ions on Nov. 1.

Bibliography Type:	Description: (Last Updated: 11/13/2018)
Abstracts for Journals and Proceedings	Held KD, Lumpkins S, Yang H, Magpayo N, Schuemann J. "LET dependence of response of irradiated and bystander cells to very low fluences of charged particles in 2D and 3D." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. Talk presented by KD Held. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012
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